

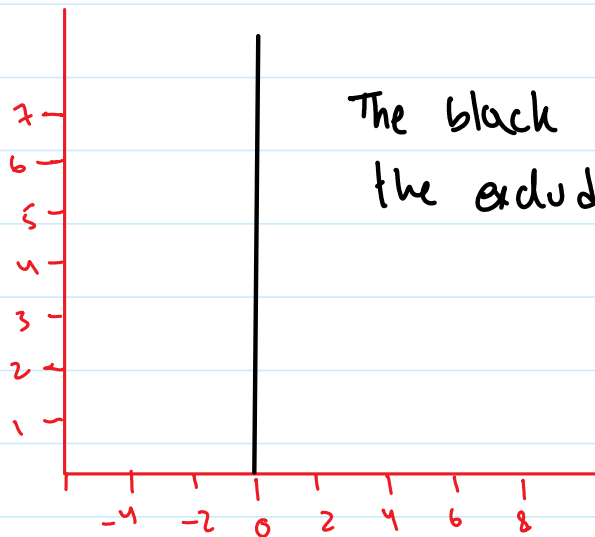
CALC HW 4

Wednesday, March 26, 2025 1:57 PM

SECTION 14.1:-

10. $f(x, y) = \sin \frac{y}{x}$

The function includes y/x where $x \neq 0$ that means, the domain includes all the real values of x, y but doesn't include the y -axis (ie: $x=0$).



26. $f(x, y) = \frac{1}{x^2 + y^2 + 1}$

$$\text{Domain} = \{(x, y) \in \mathbb{R}^2\}$$

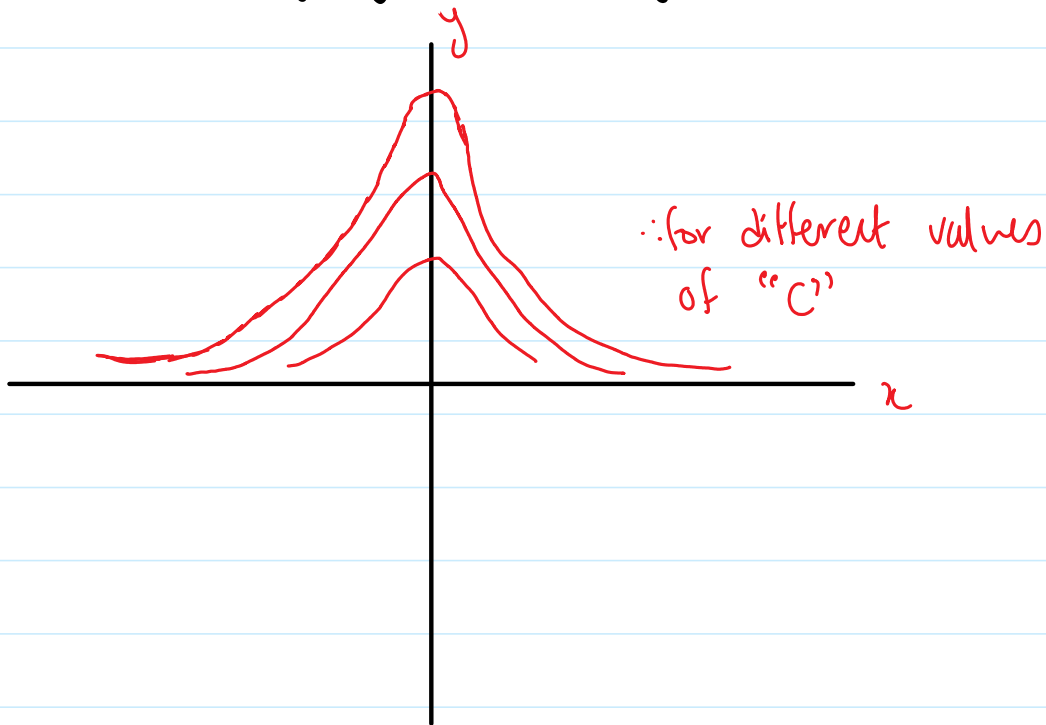
$$\text{max value} = f(0, 0) = \underline{1} = 1$$

$$0^2 + 0^2 + 1$$

- Horizontal trace:-

$$f(x, c) = \frac{1}{x^2 + c^2 + 1}$$

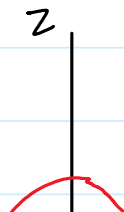
downward opening graph in xy -plane



- Vertical trace:- (for $x=c$)

$$f(c, y) = \frac{1}{c^2 + y^2 + 1}$$

- The graph will be same as horizontal trace but in xz plane:-





38. Use the contour map in Figure 25 to calculate the average rate of change:

(a) from A to B.

(b) from A to C.

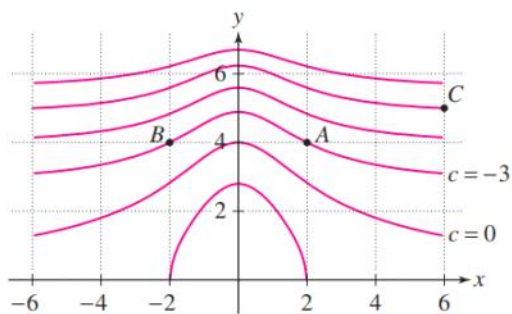


FIGURE 25

average rate of change = $\frac{\Delta \text{altitude}}{\Delta \text{horizontal.}}$

a) A $\rightarrow$ B :-

$$\text{avg} = \frac{0}{4} = 0$$

b) $A \rightarrow C$:-

$$\text{avg} = \frac{1}{6-2} = \frac{1}{4} = 0.25$$

In Exercises 44–47, $\rho(S, T)$ is seawater density (kilograms per cubic meter) as a function of salinity S (parts per thousand) and temperature T (degrees Celsius). Refer to the contour map in Figure 27.

44. Calculate the average rate of change of ρ with respect to T from B to A .

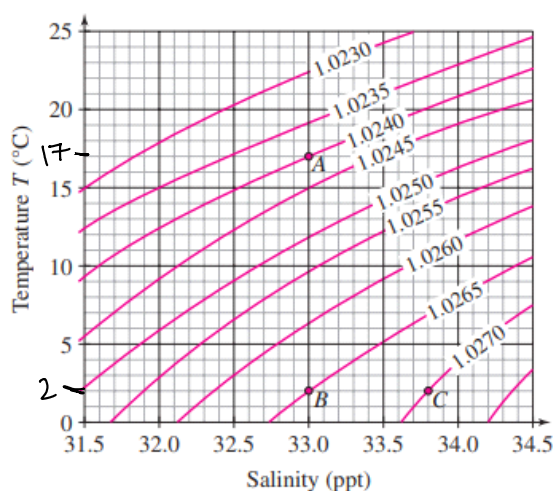


FIGURE 27 Contour map of seawater density $\rho(S, T)$ (kilograms per cubic meter).

$$\text{avg rate of change} = \frac{\Delta \rho}{\Delta T} = \frac{1.0265 - 1.0240}{2 - 17} = -1.66 \times 10^{-4} \text{ kgm}^3 / ^\circ\text{C}$$

53. Let temperature in 3-space be given by $T(x, y, z) = \frac{x^2}{4} + \frac{y^2}{9} + z^2$. Draw isotherms corresponding to temperatures $T = 0, 1, 4$.

for $t=0$:-

$$\frac{x^2}{9} + \frac{y^2}{9} + z^2 = 0$$

that means $x=0$, $y=0$, $z=0$
 $(0,0,0)$

level surface of T is at a single point.

for $t=1$:-

$$\frac{x^2}{9} + \frac{y^2}{9} + z^2 = 1$$

$a=2$ along x -axis

$b=3$ along y -axis

$c=1$ along z -axis

This represents an ellipsoid centered at the origin with
 semi-axes a, b, c

for $t=4$:-

$$\frac{x^2}{9} + \frac{y^2}{9} + z^2 = 4$$

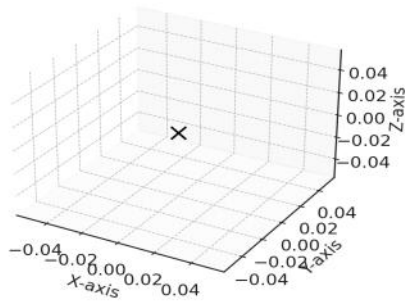
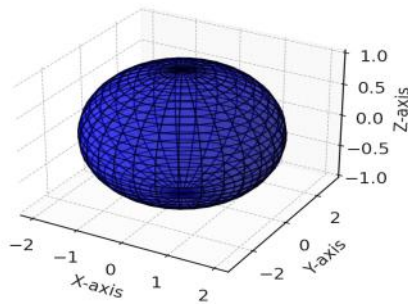
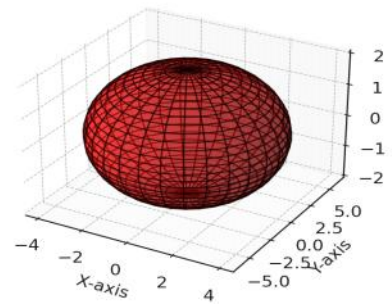
$a=4$

$b=6$

$c=2$

- This represents a larger ellipsoid with semi-axes at a, b, c .

Plotting isotherms is really difficult with hand; using calculator

Isothermal Surface for $T = 0$ Isothermal Surface for $T = 1$ Isothermal Surface for $T = 4$ 

57. Let

$$f(x, y) = \frac{x}{\sqrt{x^2 + y^2}} \quad \text{for } (x, y) \neq (0, 0)$$

Write f as a function $f(r, \theta)$ in polar coordinates, and use this to find the level curves of f .

Step 1:-

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$f(x, y) = \frac{x}{\sqrt{x^2 + y^2}} = \frac{r \cos \theta}{r} = \cos \theta$$

$$f(r, \theta) = \cos \theta$$

Step 2:-

level curve = constant

$$f(r, \theta) = \cos \theta = \text{constant}$$

$$\cos \theta = C$$

- As $\cos \theta$ is not dependent upon " r " we can use any value of r except at origin.
- Since $\cos \theta$ is only dependent on angle " θ ", hence the level curves are the line passing through the origin.

SECTION 14.2

18. Let $f(x, y) = x^3/(x^2 + y^2)$ and $g(x, y) = x^2/(x^2 + y^2)$. Using polar coordinates, prove that

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$$

and that $\lim_{(x,y) \rightarrow (0,0)} g(x, y)$ does not exist. *Hint:* Show that $g(x, y) = \cos^2 \theta$ and observe that $\cos \theta$ can take on any value between -1 and 1 as $(x, y) \rightarrow (0, 0)$.

$$f(x, y) = \frac{x^3}{x^2 + y^2}$$

$$f(r, \theta) = \frac{r^3 \cos^3 \theta}{r^2} = r \cos^3 \theta$$

$$\text{for } \lim_{(x,y) \rightarrow (0,0)}$$

$$r^2 = x^2 + y^2 = 0^2 + 0^2 = 0$$

$$\boxed{r=0}$$

put the value of 'r' in equation (i):-

$$f(x, y) = r \cos^3 \theta = 0 \times \cos^3 \theta = \boxed{0}$$

Hence proved!

$$\bullet \quad g(x, y) = \frac{x^2}{x^2 + y^2}$$

$$g(r, \theta) = \frac{\cancel{x^2} \cos^2 \theta}{\cancel{x^2}} = \cos^2 \theta.$$

Here, the function 'g' is only dependent on " θ " and not on 'r', that means we can take any value between '0' to '1' and it depends upon the direction, we will get different answers for different solutions hence the limit does not exist.

26. Evaluate $\lim_{(x, y) \rightarrow (0, 0)} \tan x \sin \left(\frac{1}{|x| + |y|} \right).$

$\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$, we need to determine that whether the limit exist or not.

- Approach Along the x -axis :- (let $y=0$)

$$f(x,0) = \tan x \cdot \sin\left(\frac{1}{|x|}\right)$$

As $x \rightarrow 0$, $\tan x \rightarrow 0$, but the term $\sin\left(\frac{1}{|x|}\right)$ oscillates between $(-1, 1)$

which means:-

$$f(x,0) = 0 \times \sin\left(\frac{1}{x}\right)$$

when $x \rightarrow 0$

$$\lim_{x \rightarrow 0} f(x,0) = 0$$

- Approach along y -axis:- let $x=0$

$$f(0,y) = \tan 0 \cdot \sin\left(\frac{1}{|y|}\right) = 0 \cdot \sin\left(\frac{1}{|y|}\right) = 0$$

$$\lim_{y \rightarrow 0} f(0,y) = 0$$

Hence the limit exist at origin

37. $\lim_{(x,y) \rightarrow (0,0)} \tan(x^2 + y^2) \tan^{-1}\left(\frac{1}{x^2 + y^2}\right)$

$$f(x, y) = \tan(x^2 + y^2) \cdot \tan^{-1}\left(\frac{1}{x^2 + y^2}\right)$$

as $x \rightarrow 0$ and $y \rightarrow 0$; $x^2 + y^2 = 0$

$$f(0, 0) = \tan(0) \cdot \tan^{-1}\left(\frac{1}{0}\right)$$

$$= \tan(0) \cdot \tan^{-1}(\infty) = 0 \times \pi/2 = 0$$

- Hence the limit exists and is equal to zero.

41. Let $f(x, y) = \frac{x^3 + y^3}{x^2 + y^2}$.

(a) Show that

$$|x^3| \leq |x|(x^2 + y^2), \quad |y^3| \leq |y|(x^2 + y^2)$$

(b) Show that $|f(x, y)| \leq |x| + |y|$.

(c) Use the Squeeze Theorem to prove that $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$.

a)

- we know that,

$$x^2 \leq x^2 + y^2$$

we can factor x^3 as $|x^2| |x|$

$$|x^3| = |x| |x^2| \leq |x| (x^2 + y^2)$$

$x^2 + y^2 \geq x^2$ holds because, x^3 does not exceed $|x|(x^2 + y^2)$.

• for y^3 :-

$$|y^3| = |y| |y^2| \leq |y| (x^2 + y^2)$$

b) Using equations from part (a):-
prove $(f(x, y)) \leq |x| + |y|$

$$|x^3| + |y^3| \leq |x| (x^2 + y^2) + |y| (x^2 + y^2)$$

$$x^3 + y^3 \leq (x^2 + y^2) (|x| + |y|)$$

$$\frac{x^3 + y^3}{x^2 + y^2} \leq |x| + |y|$$

as we know that $f(x, y) = \frac{x^3 + y^3}{x^2 + y^2}$

$$f(x, y) \leq |x| + |y| \quad \text{Hence proved.}$$

c) from part b $\rightarrow$ we have $f(x, y) \leq |x| + |y|$

So,

$$\lim_{(x, y) \rightarrow (0, 0)} (|x| + |y|) = 0 + 0 = 0$$

limit exists, and is equal to zero.

42. Let $a, b \geq 0$. Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^a y^b}{x^2 + y^2} = 0$ if $a + b > 2$ and that the limit does not exist if $a + b \leq 2$.

convert to polar curves:-

$$x^a = r^a \cos^a \theta$$

$$y^b = r^b \sin^b \theta$$

$$= \lim_{x,y \rightarrow 0,0} \frac{r^a \cos^a \theta \cdot r^b \sin^b \theta}{r^2 \cos^2 \theta + r^2 \sin^2 \theta}$$

$$= \frac{r^{a+b} \cos^a \theta \cdot \sin^b \theta}{r^2 (\cos^2 \theta + \sin^2 \theta)} \quad \therefore \cos^2 \theta + \sin^2 \theta = 1$$

$$= r^{a+b-2} \cos^a \theta \sin^b \theta$$

• for $a+b > 2$:-

as $a+b > 2$ so $r^{a+b-2} \rightarrow 0$ as $r \rightarrow 0$, since $\cos^a \theta \cdot \sin^b \theta$ is bounded so,

$$= 0 \times \cos^a \theta \sin^b \theta = 0$$

$$\text{Hence the limit } \lim_{x,y \rightarrow 0,0} \frac{x^a y^b}{x^2 + y^2} = 0 \quad \text{if } a+b > 2$$

• for $a+b < 2$:-

as $a+b < 2$ so $r^{a+b-2} \rightarrow \infty$ as $r \rightarrow 0$, so the function blows up:-

$$\lim_{x,y \rightarrow 0,0} \frac{x^a y^b}{x^2 + y^2} \text{ if } a+b < 2 \text{ the limit does not exist.}$$

SECTION 14.3:-

$$40. w = \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

Partial differentiation wrt x :-

$$f = x \quad f' = 1$$

$$g(x) = (x^2 + y^2 + z^2)^{3/2}$$

$$g'(x) = \frac{3}{2} (x^2 + y^2 + z^2)^{1/2} \cdot 2x = 3x(x^2 + y^2 + z^2)^{1/2}$$

applying quotient rule-

$$\frac{\partial w}{\partial x} = \frac{f'g - g'f}{g^2} = \frac{(x^2 + y^2 + z^2)^{3/2} - 3x^2(x^2 + y^2 + z^2)^{1/2}}{(x^2 + y^2 + z^2)^3}$$

$$\frac{\partial w}{\partial x} = \frac{(x^2 + y^2 + z^2)^{1/2} [(x^2 + y^2 + z^2) - 3x^2]}{(x^2 + y^2 + z^2)^3}$$

$$\frac{\partial w}{\partial x} = \frac{y^2 + z^2 - 2x^2}{(x^2 + y^2 + z^2)^{5/2}} \text{ Ans.}$$

- Partial differentiation wrt y :-

$$\frac{\partial w}{\partial y} = \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$f = x \quad f' = 0$$

$$g = (x^2 + y^2 + z^2)^{3/2} \quad g' = \frac{3}{2} (x^2 + y^2 + z^2)^{1/2} y$$

$$= 3y(x^2 + y^2 + z^2)$$

$$\frac{\partial w}{\partial y} = \frac{f'g - g'f}{g^2} = \frac{-3y(x^2 + y^2 + z^2)^{1/2} \cdot x}{(x^2 + y^2 + z^2)^3}$$

$$= \frac{-3xy}{(x^2 + y^2 + z^2)^{5/2}} \quad \text{Ans.}$$

- Partial differentiation wrt z :-

$$w = \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\frac{\partial w}{\partial z} = \quad f = x$$

$$f' = 0$$

$$g = (x^2 + y^2 + z^2)^{3/2}$$

$$g' = 3z(x^2 + y^2 + z^2)^{1/2}$$

$$\frac{\partial w}{\partial z} = \frac{f'g - g'f}{g^2} = \frac{-3z(x^2 + y^2 + z^2)^{1/2} \cdot x}{(x^2 + y^2 + z^2)^3}$$

$$= \frac{-3xz}{(x^2 + y^2 + z^2)^{5/2}} \quad \text{Ans}$$

$$= \frac{-3xz}{(x^2 + y^2 + z^2)^{5/2}} \text{ Ans}$$

48. The **wind-chill temperature** W measures how cold people feel (based on the rate of heat loss from exposed skin) when the outside temperature is $T^\circ\text{C}$ (with $T \leq 10$) and wind velocity is v m/s (with $v \geq 2$):

$$W = 13.1267 + 0.6215T - 13.947v^{0.16} + 0.486Tv^{0.16}$$

Calculate $\partial W / \partial v$ at $(T, v) = (-10, 15)$ and use this value to estimate ΔW if $\Delta v = 2$.

• Computing $\frac{\partial W}{\partial v}$

$$W = 13.1267 + 0.6215T - 13.947v^{0.16} + 0.486Tv^{0.16}$$

$$\frac{\partial W}{\partial v} = -0.16v^{-0.84}(-13.947 + 0.486T)$$

• $(T, v) = (-10, 15)$

$$\frac{\partial W}{\partial v} = 0.16(15^{-0.84})(-13.947 + 0.486(-10))$$

$$= 0.16(15^{-0.84})(-13.947 - 4.86)$$

$$= -0.193$$

• estimating ΔW with $\Delta v = 2$:-

$$\Delta W = \frac{\partial W}{\partial v} \cdot \Delta v = (-0.193) \times 2 = -0.386$$

The temperature decreased by 0.386°C

49. The volume of a right-circular cone of radius r and height h is $V = \frac{\pi}{3}r^2h$. Suppose that $r = h = 12$ cm. What leads to a greater increase in V , a 1-cm increase in r or a 1-cm increase in h ? Argue using partial derivatives.

$$\frac{\partial V}{\partial r} = \frac{\pi}{3} (2rh) = \frac{2\pi}{3} rh$$

$$\frac{\partial V}{\partial h} = \frac{\pi}{3} r^2$$

• for $r=h=12$:-

$$\frac{\partial V}{\partial r} = \frac{2\pi}{3} (12)(12) = 96\pi \approx 301.59$$

$$\frac{\partial V}{\partial h} = \frac{\pi}{3} (12)^2 = 48\pi = 150.79$$

• for $r=h=13$:-

$$\frac{\partial V}{\partial r} = \frac{2\pi}{3} (13)(13) = 353.95$$

$$\frac{\partial V}{\partial h} = \frac{\pi}{3} (13)^2 = 176.97$$

$$\Delta V \text{ by } r = 301.59 - 353.95 = -52.36$$

$$n = 150.79 - 176.97 = -26.18$$

- as ΔV increases more with respect to 'r'.

50. Use the linear approximation to estimate the percentage change in volume of a right-circular cone of radius $r = 40$ cm if the height is increased from 40 to 41 cm.

- Volume of right circular cone $= \frac{\pi}{3} r^2 h$

$$\frac{\partial V}{\partial h} = \frac{\pi}{3} r^2$$

- substituting $r = 40$

$$\frac{\partial V}{\partial h} = \frac{\pi}{3} (40)^2 = \frac{1600\pi}{3} = 1675.51$$

- estimating the change in volume-

$$\Delta V \approx \frac{\partial V}{\partial h} \cdot \Delta h$$

given that $\Delta h = 41 - 40 = 1$

$$\Delta V \approx \frac{\pi}{3} (41)^2 = 1760.33$$

- $i. = \frac{1675.51}{1760.33} \times 100 = 95.18 \%$

= 4.81% change

73. Find a function such that $\frac{\partial f}{\partial x} = 2xy$ and $\frac{\partial f}{\partial y} = x^2$.

- integrate $\frac{\partial f}{\partial x}$ wrt 'x':-

$$\int \frac{\partial f}{\partial x} = 2y \int x \, dx = \frac{x^2}{2} \cdot (2y) = \boxed{x^2 y}$$

- integrate $\frac{\partial f}{\partial y}$ wrt 'y':-

$$\int \frac{\partial f}{\partial y} = \int x^2 \cdot dy = \boxed{x^2 y}$$

The function is $x^2 y$.

78. The function

$$f(x, t) = \frac{1}{2\sqrt{\pi t}} e^{-x^2/4t}$$

describes the temperature profile along a metal rod at time $t > 0$ when a burst of heat is applied at the origin (see Example 11). A small bug sitting on the rod at distance x from the origin feels the temperature rise and fall as heat diffuses through the bar. Show that the bug feels the maximum temperature at time $t = \frac{1}{2}x^2$.

differentiating wrt 't':-

1.1

2.1

$$g(t) = \frac{1}{2\sqrt{\pi t}} \quad h(t) = e^{-x^2/4t}$$

using product rule:-

$$= g'h + h'g$$

$$\frac{\partial f}{\partial t} = \left(-\frac{1}{4\sqrt{\pi}} t^{-3/2} \right) h(t) + \left(\frac{1}{2\sqrt{\pi}} t^{-1/2} \right) \left(h(t) \frac{x^2}{4t^2} \right)$$

$$\frac{\partial f}{\partial t} = \frac{h(t)}{\sqrt{\pi}} \left[-\frac{1}{4} t^{-3/2} + \frac{x^2}{8} t^{-5/2} \right]$$

set the derivative = 0

$$-\frac{1}{4} t^{-3/2} + \frac{x^2}{8} t^{-5/2} = 0$$

$$\frac{1}{4} t^{-3/2} = \frac{x^2}{8} t^{-5/2}$$

$$2 t^{-3/2} = x^2 t^{-5/2}$$

$$2 t^{-3/2+5/2} = x^2$$

$$2t = x^2$$

$$\boxed{t = \frac{x^2}{2}}$$

SECTION 14.4

In Exercises 3–10, find an equation of the tangent plane at the given point.

10. $f(x, y) = \ln(4x^2 - y^2), \quad (1, 1)$

$$[a, b] = [1, 1]$$

Equation of tangent:-

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

$$f_x(x, y) = \frac{1}{4x^2 - y^2} \cdot 8x = \frac{8x}{4x^2 - y^2}$$

$$f_x(a, b) = \frac{8(1)}{4(1) - 1} = \frac{8}{3}$$

$$f_y(x, y) = \frac{1}{4x^2 - y^2} \cdot -2y = \frac{-2y}{4x^2 - y^2}$$

$$f_y(a, b) = \frac{-2(1)}{4 - 1} = \frac{-2}{3}$$

Now,

$$f(a, b) = \ln(4x^2 - y^2) = \ln(4 - 1) = \ln(3)$$

$$z = \ln(3) + \left(\frac{8}{3}\right)(x-1) + \left(\frac{-2}{3}\right)(y-1)$$

$$\boxed{z = \ln(3) + \frac{8}{3}(x-1) - \frac{2}{3}(y-1)} \text{ Ans.}$$

18. Let $f(x, y) = x^2/(y^2 + 1)$. Use the linear approximation at an appropriate point (a, b) to estimate $f(4.01, 0.98)$.

$$f(a + \Delta x, b + \Delta y) = f(a, b) + f_x(a, b)\Delta x + f_y(a, b)\Delta y$$

let take (a, b) as $(4, 1)$ as it will make solution simpler.

$$f(a, b) = \frac{x^2}{y^2 + 1} = \frac{4^2}{2} = \frac{16}{2} = 8$$

$$f_x(x, y) = \frac{2x}{y^2 + 1}$$

$$f_x(a, b) = \frac{2(4)}{2} = \frac{8}{2} = 4$$

$$f_y(x, y) = \frac{x^2}{y^2 + 1} = \frac{-x^2}{(y^2 + 1)^2} \cdot 2y = \frac{-2x^2 y}{y^2 + 1}$$

$$f_y(a, b) = \frac{-2(4)^2(1)}{(1+1)^2} = \frac{-16}{2} = -8$$

$$= \frac{1}{(1+1)^2} = \frac{1}{2} = 0.5$$

Now,

$$\Delta f = 8 + 4(4.01 - 4) + (-8)(0.98 - 1)$$

$$\Delta f = 8 + 0.2 = \boxed{8.2}$$

30. In the derivation of the equation for the tangent plane that appears in Theorem 1, we chose $\mathbf{n} = \mathbf{v} \times \mathbf{u}$ for the normal vector to the plane. How would the choice of $\mathbf{n} = \mathbf{u} \times \mathbf{v}$ for the normal vector have affected the resultant equation?

- as we know that cross product is not commutative
so

$$\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$$

- however using $\mathbf{u} \times \mathbf{v}$ for normal vector will not affect the normal vector, but only its direction will be opposite.

35. A cylinder of radius r and height h has volume $V = \pi r^2 h$.

(a) Use the linear approximation to show that

$$\frac{\Delta V}{V} \approx \frac{2\Delta r}{r} + \frac{\Delta h}{h}$$

(b) Estimate the percentage increase in V if r and h are each increased by 2%.

(c) The volume of a certain cylinder V is determined by measuring r and h . Which will lead to a greater error in V : a 1% error in r or a 1% error in h ?

a)

$$dV = \frac{\partial V}{\partial r} \cdot dr + \frac{\partial V}{\partial h} \cdot dh$$

$$\frac{\partial V}{\partial r} = 2\pi rh \quad \frac{\partial V}{\partial h} = \pi r^2$$

$$dV = (2\pi rh) \cdot dr + (\pi r^2) \cdot dh$$

dividing by $\pi r^2 h$ on b.s:-

$$\frac{dV}{V} = \frac{(2\pi rh) dr}{\pi r^2 h} + \frac{\pi r^2}{\pi r^2 h} \cdot dh$$

$$\boxed{\frac{\Delta V}{V} = \frac{2 \Delta r}{r} + \frac{\Delta h}{h}}$$

where $\Delta r = dr$
 $\Delta h = dh$

b)

$$\frac{\Delta V}{V} = \frac{2 \Delta r}{r} + \frac{\Delta h}{h}$$

so,

$$\frac{\Delta r}{r} = 2\% \quad \text{and} \quad \frac{\Delta h}{h} = 2\%$$

$$\frac{\Delta V}{V} = 2(2\%) + 2\% = 4 + 2 = \boxed{6\%}$$

The volume is increased by 6%.

c) $\frac{Dr}{r} = 1\% \text{ error}$

$$DV = 2(1\%) = 2\% \text{ error}$$

$$\frac{Dh}{h} = 1\%$$

$$DV = 1\%$$

That means, 1% error in r will lead to a greater error.

37. The monthly payment for a home loan is given by a function $f(P, r, N)$, where P is the principal (initial size of the loan), r the interest rate, and N the length of the loan in months. Interest rates are expressed as a decimal: A 6% interest rate is denoted by $r = 0.06$. If $P = \$100,000$, $r = 0.06$, and $N = 240$ (a 20-year loan), then the monthly payment is $f(100,000, 0.06, 240) = 716.43$. Furthermore, at these values, we have

$$\frac{\partial f}{\partial P} = 0.0071, \quad \frac{\partial f}{\partial r} = 5769, \quad \frac{\partial f}{\partial N} = -1.5467$$

Estimate:

(a) The change in monthly payment per \$1000 increase in loan principal

(b) The change in monthly payment if the interest rate increases to $r = 6.5\%$ and $r = 7\%$

(c) The change in monthly payment if the length of the loan increases to 24 years

a) Change in P :- $\Delta P = 1000$

$$\Delta f \approx \frac{\partial f}{\partial P} \cdot \Delta P = (0.0071)(1000) = 7.1$$

The monthly payment is increased by 7.10\$ per 1000% increase in loan principal.

b) if monthly payment interest is increased to 6.5%.

$$\Delta r = 0.065 - 0.06 = 0.005$$

$$\Delta f = (5769)(0.005) = 28.85$$

for $r = 7\%$:-

$$\Delta f = (5769)(0.01) = 57.69$$

- \$28.85 when r increased to 6.5%.
- \$57.69 when r increased to 7%.

c) Change in monthly payments:-

$$\Delta N = 288 - 240 = 48$$

$$\Delta f = \frac{\partial f}{\partial N} \cdot \Delta N = (-1.546)(48) = -74.24$$

- The monthly payment is decreased by \$74.24.

38. Automobile traffic passes a point P on a road of width w feet (ft) at an average rate of R vehicles per second. Although the arrival of automobiles is irregular, traffic engineers have found that the average waiting time T until there is a gap in traffic of at least t seconds is approximately $T = te^{Rt}$ seconds (s). A pedestrian walking at a speed of 3.5 ft/s (5.1 mph) requires $t = w/3.5$ s to cross the road. Therefore, the average time the pedestrian will have to wait before crossing is $f(w, R) = (w/3.5)e^{wR/3.5}$ s.

(a) What is the pedestrian's average waiting time if $w = 25$ ft and $R = 0.2$ vehicle per second?

(b) Use the linear approximation to estimate the increase in waiting time if w is increased to 27 ft.

(c) Estimate the waiting time if the width is increased to 27 ft and R decreases to 0.18.

(d) What is the rate of increase in waiting time per 1-ft increase in width when $w = 30$ ft and $R = 0.3$ vehicle per second?

$$\begin{aligned} \text{a)} \quad f(25, 0.2) &= \frac{25}{3.5} e^{\frac{25 \cdot 0.2}{3.5}} = 7.174 \cdot e^{1.428} \\ &= 29.8 \text{ seconds} \end{aligned}$$

b) using linear approximation:-

$$\Delta f = \frac{\partial f}{\partial w} \cdot \Delta w$$

$$\frac{\partial f}{\partial w} = \frac{1}{3.5} e^{wR/3.5} + \frac{w}{3.5} e^{wR/3.5} \cdot \frac{R}{3.5}$$

$$\frac{\partial f}{\partial w} = \frac{1}{3.5} e^{wR/3.5} + \frac{w}{3.5} e^{wR/3.5} \cdot \frac{R}{3.5}$$

$$\frac{\partial f}{\partial w} = \frac{e^{wR/3.5}}{3.5} \left(1 + \frac{wR}{3.5} \right)$$

At $w=25, R=0.2$:

$$\frac{\partial f}{\partial w} = \frac{4.171}{3.5} (1 + 1.428) = 1.191 \times 2.42 = 2.894$$

$$\Delta w = 2 \Rightarrow \Delta f = 5.79 \text{ seconds}$$

c)

$$f(27, 0.18) = \frac{27}{3.5} e^{\frac{27 \cdot 0.18}{3.5}} = 7.7143 \cdot e^{1.388}$$

$$= 7.714 \times 4.009$$

$$= 30.93 \text{ seconds}$$

d) $w=30, R=0.3$:-

$$\frac{\partial f}{\partial w} = \frac{e^{wR/3.5}}{3.5} \left(1 + \frac{wR}{3.5} \right)$$

$$\frac{\partial f}{\partial w} = \frac{13.096}{3.5} \times 3.5714 = 13.36 \text{ sec/ft}$$

8. $r(x, y, z, w) = xze^{yw}$

The gradient is given by:-

$$\nabla r = \left(\frac{\partial r}{\partial x}, \frac{\partial r}{\partial y}, \frac{\partial r}{\partial z}, \frac{\partial r}{\partial w} \right)$$

- Compute the partial derivatives:-
- $\frac{\partial r}{\partial x} = ze^{yw}$
- $\frac{\partial r}{\partial y} = xzwe^{yw}$
- $\frac{\partial r}{\partial z} = xe^{yw}$
- $\frac{\partial r}{\partial w} = x y z e^{yw}$
- Gradient vector = $\nabla r = \left(ze^{yw}, xzwe^{yw}, xe^{yw}, x y z e^{yw} \right)$

14. $f(x, y) = \cos(y - x)$, $\mathbf{r}(t) = \langle e^t, e^{2t} \rangle$, $t = \ln 3$

- Applying chain rule :-

$$\frac{d}{dt} f(\mathbf{r}(t)) = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

$$\frac{d}{dt} f(r(t)) = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt}$$

- $\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} \cos(y-x) = -(-\sin(y-x)) = \sin(y-x)$

- $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \cos(y-x) = -\sin(y-x)$

- Now,

$$x(t) = e^t, \quad dx/dt = e^t$$

$$y(t) = e^{2t}, \quad dy/dt = 2e^{2t}$$

- Put all the values in chain rule:-

$$\frac{d}{dt} f(r(t)) = \sin(y-x) \cdot e^t + (-\sin(y-x)) 2e^{2t}$$

- substitute x, y in equation:-

$$\frac{d}{dt} f(r(t)) = \sin(e^{2t} - e^t) \cdot e^t - 2 \sin(e^{2t} - e^t) \cdot e^{2t}$$

- evaluate:- $t = \ln(3)$

- $x(\ln(3)) = e^{\ln 3} = 3$

- $y(\ln(3)) = e^{2\ln 3} = 9$

$$1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad 7 \quad 8 \quad 9 \quad 10$$

$$\frac{d}{dt} f(r(\ln 3)) = \sin(9-3) \cdot 3 - 2 \sin(9-3) \cdot 9$$

$$\boxed{\frac{d}{dt} (f(r(t))) = -15 (\sin(6))} \text{ Ans}$$

In Exercises 21–30, calculate the directional derivative in the direction of $\mathbf{v}$ at the given point. Remember to normalize the direction vector.

30. $g(x, y, z) = x \ln(y + z)$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $P = (2, e, e)$

- find the gradient:-

$$\nabla g = \left\langle \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}, \frac{\partial g}{\partial z} \right\rangle$$

- $\frac{\partial g}{\partial x} = \ln(y+z)$

- $\frac{\partial g}{\partial y} = \frac{x}{y+z}$

- $\frac{\partial g}{\partial z} = \frac{x}{y+z}$

$$\nabla g = \left\langle \ln(y+z), \frac{x}{y+z}, \frac{x}{y+z} \right\rangle$$

- substitute $x=2$,
 $y=e$,
 $z=e$.

$$\nabla g(2, e, e) = \left\langle \ln(2e), \frac{2}{2e}, \frac{2}{2e} \right\rangle$$

$$\nabla g(2, e, e) = \left\langle \ln(2e), 1/e, 1/e \right\rangle$$

- Normalize the directional vector v :-

$$|v| = \sqrt{2^2 + (-1)^2 + (1)^2} = \sqrt{6}$$

$$u = \frac{v}{|v|} = \frac{1}{\sqrt{6}} (2, -1, 1) = \left\langle \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right\rangle$$

- Directional derivative =

$$D_u g = \nabla g(2, e, e) \cdot u$$

$$= \frac{2 \ln(2e)}{\sqrt{6}} - \frac{1}{e\sqrt{6}} + \frac{1}{e\sqrt{6}}$$

$$\boxed{D_u g = \frac{2 \ln(2e)}{\sqrt{6}}} \quad \text{Ans.}$$

32. Find the directional derivative of $f(x, y, z) = xy + z^3$ at the point $P = (3, -2, -1)$ in the direction pointing to the origin.

$$\nabla f = \left\langle \underline{\frac{\partial f}{\partial x}}, \underline{\frac{\partial f}{\partial y}}, \underline{\frac{\partial f}{\partial z}} \right\rangle$$

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

- $\frac{\partial f}{\partial x} = y$

- $\frac{\partial f}{\partial y} = x$

- $\frac{\partial f}{\partial z} = 3z^2$

$$\nabla f = \langle y, x, 3z^2 \rangle$$

- Put the value of x, y, z :-

$$\begin{aligned} \nabla f(3, -2, -1) &= \langle -2, 3, 3(-1)^2 \rangle \\ &= \langle -2, 3, 3 \rangle \end{aligned}$$

- as it is pointing towards origin ie. $(0, 0, 0)$

$$V = \langle 0-3, 0-(-2), 0-(-1) \rangle = \langle -3, 2, 1 \rangle$$

- finding normal :-

$$|V| = \sqrt{3^2 + 2^2 + 1^2} = \sqrt{14}$$

Now,

$$u = \frac{V}{|V|} = \left\langle \frac{-3}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}} \right\rangle$$

- Calculate the directional derivative:-

$$D_u f = \nabla f(3, -2, -1) \cdot u$$

$$D_u f = \langle -2, 3, 3 \rangle \cdot \left\langle \frac{-3}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{1}{\sqrt{14}} \right\rangle$$

$$D_u f = \frac{6}{\sqrt{14}} + \frac{6}{\sqrt{14}} + \frac{3}{\sqrt{14}}$$

$$D_u f = \frac{15}{\sqrt{14}}$$

38. Let $f(x, y) = xe^{x^2-y}$ and $P = (1, 1)$.

(a) Calculate $\|\nabla f_P\|$.

(b) Find the rate of change of f in the direction ∇f_P .

(c) Find the rate of change of f in the direction of a vector making an angle of 45° with ∇f_P .

a)

$$\frac{\partial f}{\partial x} = e^{x^2-y} + x(2x)e^{x^2-y} = (1+2x^2)e^{x^2-y}$$

$$\frac{\partial f}{\partial y} = x(-1)e^{x^2-y} = -xe^{x^2-y}$$

$$\nabla f(x, y) = \left\langle (1+2x^2)e^{x^2-y}, -xe^{x^2-y} \right\rangle$$

$$\nabla f(1,1) = \langle (1+2(1))e^{-1}, -(1)e^{-1} \rangle$$

$$\nabla f(1,1) = \langle 3, -1 \rangle$$

• Now,

$$|\nabla f_p| = \sqrt{3^2 + 1^2} = \sqrt{10}$$

b) rate of change = $|\nabla f_p| = \sqrt{10}$

c) rate of change of 'f' making 45° angle with ' f_p ' :-

$$\Delta_u f = |\nabla f_p| \cos \theta$$

$$\Delta_u f = \sqrt{10} \cos(45) = \frac{2\sqrt{5}}{2} = \sqrt{5}$$

42. Find a vector normal to the surface $3z^3 + x^2y - y^2x = 1$ at $P = (1, -1, 1)$.

let

$$f(x, y, z) = 3z^3 + x^2y - y^2x$$

$$\nabla f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

• $\frac{\partial f}{\partial x} = 2xy - y^2$

r_n

$$\cdot \frac{\partial f}{\partial y} = x^2 - 2xy$$

$$\cdot \frac{\partial f}{\partial z} = 9z^2$$

$$\nabla f(x, y, z) = \langle 2xy - y^2, x^2 - 2xy, 9z^2 \rangle$$

• Substitute the values:-

$$x = 1, y = -1, z = 1$$

$$\begin{aligned}\nabla f(1, -1, 1) &= (-2 - 1, 1 + 2, 9) \\ &= \langle -3, 3, 9 \rangle\end{aligned}$$

or can also be written as

$$= \langle -1, 1, 3 \rangle$$

60. Let $f(x, y) = \tan^{-1} \frac{x}{y}$ and $\mathbf{u} = \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle$.

(a) Calculate the gradient of f .

(b) Calculate $D_{\mathbf{u}} f(1, 1)$ and $D_{\mathbf{u}} f(\sqrt{3}, 1)$.

(c) Show that the lines $y = mx$ for $m \neq 0$ are level curves for f .

(d) Verify that ∇f_P is orthogonal to the level curve through P for $P = (x, y) \neq (0, 0)$.

a) calculating gradient:-

$$\cdot \frac{\partial f}{\partial x} = \frac{1}{1 + \left(\frac{x}{y}\right)^2} \cdot \frac{1}{y} = \frac{y}{x^2 + y^2}$$

$$\cdot \frac{\partial f}{\partial y} = \frac{1}{1+(x/y)^2} \cdot \left(-\frac{x}{y^2}\right) = \frac{-x}{x^2+y^2}$$

$$\nabla f(x, y) = \left\langle \frac{y}{x^2+y^2}, \frac{x}{x^2+y^2} \right\rangle$$

at (1,1):-

$$\cdot \nabla f(1,1) = \left\langle 1/2, -1/2 \right\rangle$$

$$D_u f(1,1) = \left\langle 1/2, -1/2 \right\rangle \cdot \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle = \frac{\sqrt{2}}{4} - \frac{\sqrt{2}}{4}$$

at ($\sqrt{3}, 1$):-

$$\cdot \nabla f = \left\langle \frac{1}{\sqrt{3}^2+1^2}, -\frac{\sqrt{3}}{\sqrt{3}^2+1^2} \right\rangle = \left\langle \frac{1}{4}, -\frac{\sqrt{3}}{4} \right\rangle \quad = 0$$

$$\cdot D_u f(\sqrt{3}, 1) = \left\langle \frac{1}{4}, -\frac{\sqrt{3}}{4} \right\rangle \cdot \left\langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right\rangle = \frac{\sqrt{2}}{8} - \frac{\sqrt{6}}{8}$$

$$= \frac{\sqrt{2} - \sqrt{6}}{8}$$

c) let $y = mx$

$$f(x, mx) = \tan^{-1}\left(\frac{x}{mx}\right) = \tan^{-1}\left(\frac{1}{m}\right)$$

• Since $1/m$ is constant so, $\tan^{-1}(1/m)$ will also be constant:-

$$f(x, mx) = C$$

- where line $y = mx$ shows that it is the level curve for 'f'.

d) we know that $m = \frac{y_0}{x_0}$ from $y_0 = mx_0$

$$\nabla f(x_0, y_0) = \left\langle \frac{y_0}{x_0^2 + y_0^2}, \frac{-x_0}{x_0^2 + y_0^2} \right\rangle$$

$$\text{slope} = y/x$$

$$\frac{\frac{-x_0}{x_0^2 + y_0^2}}{\frac{y_0}{x_0^2 + y_0^2}} = -\frac{x_0}{y_0}$$

$$m \left(\frac{-x_0}{y_0} \right) = \left(\frac{\cancel{y_0}}{\cancel{x_0}} \right) \left(\frac{-\cancel{x_0}}{\cancel{y_0}} \right) = -1$$

- Since the product of slope is -1, the gradient is orthogonal to the level curve through 'p'.

66. Prove the Product Rule for Gradients (Theorem 1).

$$\begin{aligned} \nabla f(q) &= \left\langle \frac{\partial f(q)}{\partial x_1}, \frac{\partial f(q)}{\partial x_2}, \frac{\partial f(q)}{\partial x_3} \dots \frac{\partial f(q)}{\partial x_n} \right\rangle \\ &= f \nabla q + q \nabla f \end{aligned}$$

- lets look at the single component of $\frac{\partial fg}{\partial x_i}$

lets say i-th component.

$$\frac{\partial (fg)}{\partial x_i} = f \frac{\partial g}{\partial x_i} + g \frac{\partial f}{\partial x_i}$$

- The gradient for fg:-

$$\nabla fg = \left\langle f \frac{\partial g}{\partial x_1} + g \frac{\partial f}{\partial x_1}, f \frac{\partial g}{\partial x_2} + g \frac{\partial f}{\partial x_2}, \dots, f \frac{\partial g}{\partial x_n} + g \frac{\partial f}{\partial x_n} \right\rangle$$

making common-

$$\nabla fg = f \left\langle \frac{\partial g}{\partial x_1}, \frac{\partial g}{\partial x_2}, \frac{\partial g}{\partial x_3}, \dots, \frac{\partial g}{\partial x_n} \right\rangle + g \left\langle \frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right\rangle$$

so,

$$\nabla (fg) = f \nabla g + g \nabla f$$

here proved.

SECTION 14.6

In Exercises 11–16, use the Chain Rule to evaluate the partial derivative at the point specified.

- 16.** $\frac{\partial h}{\partial q}$ at $(q, r) = (3, 2)$, where $h(u, v) = ue^v$, $u = q^3$, $v = qr^2$

- $\frac{\partial h}{\partial u} = e^v$

- $\frac{\partial h}{\partial u} = e^v$
- $\frac{\partial h}{\partial v} = 4e^v$
- $\frac{\partial u}{\partial q} = 3q^2$
- $\frac{\partial v}{\partial q} = r^2$
- Substitute into chain rule formula-

$$\frac{\partial h}{\partial q} = (e^v)(3q^2) + (4e^v)(r^2)$$

$$\frac{\partial h}{\partial q} = 3q^2 e^v + 4r^2 e^v$$

- substitute $u \rightarrow v$ to $q \rightarrow r$:-

$$\frac{\partial h}{\partial q} = 3q^2 e^{q^2 r^2} + (q^3) r^2 e^{q^2 r^2}$$

- put the values $(q, r) = (3, 2)$:-

$$\frac{\partial h}{\partial q} (3, 2) = 3(3^2) e^{(3)(2)^2} + (3)^3 (2)^2 e^{(3)(2)^2}$$

$$\frac{\partial h}{\partial q} (3, 2) = 27e^{12} + 108e^{12}$$

γ_q

$$\frac{\gamma_h}{\gamma_q}(3,2) = 135 e^{12}$$

20. The Law of Cosines states that $c^2 = a^2 + b^2 - 2ab \cos \theta$, where a, b, c are the sides of a triangle and θ is the angle opposite the side of length c .

(a) Compute $\partial\theta/\partial a$, $\partial\theta/\partial b$, and $\partial\theta/\partial c$ using implicit differentiation.

(b) Suppose that $a = 10$, $b = 16$, $c = 22$. Estimate the change in θ if a and b are increased by 1 and c is increased by 2.

a)

- Differentiate both side wrt 'a':-

$$\frac{\partial}{\partial a}(c^2) = \frac{\partial}{\partial a}(a^2 + b^2 - 2ab \cos(\theta))$$

$$0 = 2a - 2b \cos \theta + 2ab \sin(\theta) \frac{\partial \theta}{\partial a}$$

$$\frac{\partial \theta}{\partial a} = \frac{b \cos \theta - a}{ab \sin \theta}$$

- Differentiate both side wrt 'b':-

$$\frac{\partial}{\partial b}(c^2) = \frac{\partial}{\partial b}(a^2 + b^2 - 2ab \cos(\theta))$$

$$0 = 2b - 2a \cos \theta - 2ab(-\sin \theta) \frac{\partial \theta}{\partial b}$$

$$0 = 2b - 2a \cos Q + 2ab \sin Q \frac{\partial Q}{\partial b}$$

$$\frac{\partial Q}{\partial b} = \frac{a \cos Q - b}{ab \sin Q}$$

- Differentiate b.s w.r.t 'c' :-

$$\frac{\partial}{\partial c} (c^2) = \frac{\partial}{\partial c} (a^2 + b^2 - 2a \cos Q)$$

$$2c = 2ab \sin Q \frac{\partial Q}{\partial c}$$

$$\frac{\partial Q}{\partial c} = \frac{2c}{2ab \sin Q} = \frac{c}{ab \sin Q}$$

b) $a=10$ $b=16$ $c=22$, $D_a=1$ $D_b=1$ $D_c=2$

$$\cos Q = \frac{a^2 + b^2 - c^2}{2ab} = \frac{-128}{320} = -0.4$$

$$\sin Q = \sqrt{1 - \cos^2 Q} = \sqrt{1 - 0.16} = \frac{\sqrt{21}}{5}$$

- compute the partial derivatives.

$$\frac{\partial Q}{\partial a} = \frac{b \cos Q - a}{ab \sin Q} = \frac{-16 \times (0.4) - 10}{160} = \frac{-41}{20}$$

$$\frac{\partial Q}{\partial a} = \frac{-b \sin Q}{a^2 \sin Q} = \frac{-16}{10 \times 16 \times \sqrt{21}/5} = \frac{-5}{8\sqrt{21}}$$

$$\frac{\partial Q}{\partial b} = \frac{a \cos Q - b}{a^2 \sin Q} = \frac{10 \times (-0.4) - 16}{160 \times \sqrt{21}/5} = \frac{-5}{8\sqrt{21}}$$

$$\frac{\partial Q}{\partial c} = \frac{c}{a^2 \sin Q} = \frac{22}{160 \times \sqrt{21}/5} = \frac{11}{16\sqrt{21}}$$

• Total change-

$$\Delta Q = \frac{\partial Q}{\partial a} \Delta a + \frac{\partial Q}{\partial b} \Delta b + \frac{\partial Q}{\partial c} \Delta c$$

$$\Delta Q = \frac{-41}{80\sqrt{21}} - \frac{5}{8\sqrt{21}} + \frac{22}{16\sqrt{21}}$$

$$\Delta Q = \frac{19}{80\sqrt{21}} \text{ rad}$$

In Exercises 27–32, calculate the partial derivative using implicit differentiation.

30. $\frac{\partial r}{\partial t}$ and $\frac{\partial t}{\partial r}$, $r^2 = te^{s/r}$

• Differentiate both side wrt t :-

$$2r \frac{\partial r}{\partial t} = e^{s/r} + t e^{s/r} \left(\frac{-s}{r^2} \right) \frac{\partial r}{\partial t}$$

$$2r \frac{\partial r}{\partial t} = e^{s/r} + t e^{s/r} \left(\frac{-s}{r^2} \right) \frac{\partial r}{\partial t}$$

$$\frac{\partial r}{\partial t} = \frac{e^{s/r}}{2r + \frac{s t e^{s/r}}{r^2}}$$

$$t = r^2 e^{-s/r}$$

$$\boxed{\frac{\partial r}{\partial t} = \frac{e^{s/r}}{2r + s}}$$

• Calculate $\frac{\partial t}{\partial r}$:

$$2r = \frac{\partial t}{\partial r} e^{s/r} - \frac{s t e^{s/r}}{r^2}$$

$$\boxed{\frac{\partial t}{\partial r} = (2r + s) e^{-s/r}}$$

38. For all $x > 0$, there is a unique value $y = r(x)$ that solves the equation $y^3 + 4xy = 16$.

(a) Show that $dy/dx = -4y/(3y^2 + 4x)$.

(b) Let $g(x) = f(x, r(x))$, where $f(x, y)$ is a function satisfying

$$f_x(1, 2) = 8, \quad f_y(1, 2) = 10$$

Use the Chain Rule to calculate $g'(1)$. Note that $r(1) = 2$ because $(x, y) = (1, 2)$ satisfies $y^3 + 4xy = 16$.

a) $\frac{d}{dx} (y^3 + 4xy) = \frac{d}{dx} (16)$

$$a) \quad \frac{d}{dx} (y^3 + 4xy) = \frac{d}{dx} (16)$$

$$3y^2 \frac{dy}{dx} + 4y + 4x \frac{dy}{dx} = 0$$

$$\boxed{\frac{dy}{dx} = \frac{-4y}{3y^2 + 4x}}$$

b)

$$g'(x) = \frac{d}{dx} f(x, r(x))$$

$$g'(x) = f_x(x, r(x)) \cdot \frac{dx}{dx} + f_y(x, r(x)) \cdot \frac{dr}{dx}$$

• find $r'(1)$:-

since $r(x) = y$ we have $r'(x) = \frac{dy}{dx}$

$$r'(x) = \frac{-4y}{3y^2 + 4x} = \frac{-4r(1)}{3(r(1))^2 + 4(1)}$$

we know that $r(1) = 2$:-

$$r'(1) = \frac{-4 \times 2}{3 \times 4 + 4} = \frac{-8}{16} = -1/2$$

• solve for $g'(1)$:-

$$g'(1) = f_x(1, r(1)) + f_y(1, r(1)) \cdot r'(1)$$

$$g'(1) = f_x(1, 2) + f_y(1, 2) \cdot r'(1)$$

$$g'(1) = 8 + 10 \left(-\frac{1}{2}\right)$$

$$g'(1) = 8 - 5 = 3$$

SECTION 14.7

6. Show that $f(x, y) = \sqrt{x^2 + y^2}$ has one critical point P and that f is nondifferentiable at P . Does f have a minimum, maximum, or saddle point at P ?

- finding the critical point.

$$f_x = \frac{\partial}{\partial x} \sqrt{x^2 + y^2} = \frac{x}{\sqrt{x^2 + y^2}}$$

$$f_y = \frac{\partial}{\partial y} \sqrt{x^2 + y^2} = \frac{y}{\sqrt{x^2 + y^2}}$$

Now set $f_x = 0$ and $f_y = 0$

$$\frac{x}{\sqrt{x^2 + y^2}} = 0 \rightarrow x = 0$$

$$\frac{y}{\sqrt{x^2 + y^2}} = 0 \rightarrow y = 0$$

- The function is continuous at P , the partial derivatives f_x & f_y exist and are continuous in a neighborhood.

At $P = (0, 0)$, partial derivatives are undefined!

$$\text{for all } (x, y) \neq (0, 0) \quad f(x, y) = \sqrt{x^2 + y^2} > 0$$

$$\text{At } P = (0, 0), \quad f(0, 0) = 0$$

This 'f' has its minimum value at 'P'
critical point = (0, 0).

In Exercises 7–23, find the critical points of the function. Then use the Second Derivative Test to determine whether they are local minima, local maxima, or saddle points (or state that the test fails).

20. $f(x, y) = (x + y) \ln(x^2 + y^2)$

- find the partial derivatives-

$$- f_x = \frac{\partial}{\partial x} [(x+y) \ln(x^2+y^2)]$$

• using product rule-

$$f_x = \ln(x^2+y^2) \cdot 1 + (x+y) \left(\frac{2x}{x^2+y^2} \right)$$

$$f_x = \ln(x^2+y^2) + \frac{2x(x+y)}{x^2+y^2}$$

$$f_y = \ln(x^2+y^2) + \frac{2y(x+y)}{x^2+y^2}$$

• Finding critical points-

$$f_x = 0 \quad f_y = 0$$

$$\ln(x^2+y^2) + \frac{2x(x+y)}{x^2+y^2} = 0 \quad \text{--- 1}$$

$$\ln(x^2+y^2) + \frac{2y(x+y)}{x^2+y^2} = 0 \quad \text{--- 2}$$

$$eq. (2) - eq. (1) :-$$

$$\frac{2x(xy) - 2y(xy)}{x^2+y^2} = 0$$

$$\frac{2(x+y)(x-y)}{x^2+y^2} = 0$$

$$(x+y)(x-y) = 0$$

either $xy = 0$

or $x-y = 0$

so,

$$y = x, \quad y = -x$$

• substitute in $q(i)$:- ($y = -x$)

$$\ln(x^2 + (-x)^2) + \frac{2x(x-x)}{x^2+x^2} = 0$$

$$\ln(2x^2) = 0$$

$$x = \pm 1/\sqrt{2}$$

Thus, critical points = $\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ & $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

• case 2:- ($y = x$)

$$\ln(x^2 + x^2) + \frac{2x(x+x)}{x^2+x^2} = 0$$

$$\ln(2x^2) + \frac{4xy}{x^2+y^2} = 0$$

$$\ln(2x^2) = -2$$

$$x^2 = \frac{1}{2e}$$

$$x = \pm \frac{1}{e\sqrt{2}}$$

So,

$$\left(\frac{1}{e\sqrt{2}}, \frac{1}{e\sqrt{2}} \right), \left(\frac{-1}{e\sqrt{2}}, \frac{-1}{e\sqrt{2}} \right)$$

• Second derivative test:-

$$f_{xx} = \frac{\partial}{\partial x} \left[\ln(x^2+y^2) + \frac{2x(xy)}{x^2+y^2} \right]$$

$$f_{xx} = \frac{2x}{x^2+y^2} + \frac{(2(2xy))(x^2+y^2) - 2x(xy)(2x)}{(x^2+y^2)^2}$$

• Now, $f_{yy} = \frac{2y}{x^2+y^2} + \frac{2(2y+x)(x^2+y^2) - 4y^2(xy)}{(x^2+y^2)^2}$

Now,

Now,

$$f_{xy} = \frac{\partial}{\partial y} \left[\ln(x^2 + y^2) + \frac{2x(x+y)}{x^2 + y^2} \right]$$

$$f_{xy} = \frac{2y}{x^2 + y^2} + \frac{(2x)(x^2 + y^2) - 2x(x+y)(2y)}{(x^2 + y^2)^2}$$

• Now for Discriminant:-

$$D = f_{xx} f_{yy} - f_{xy}^2$$

- Evaluating critical points:-

$$C = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

$$x = 1/\sqrt{2} \quad y = -1/\sqrt{2}$$

• Compute f_{xx} , f_{yy} , f_{xy} :-

$$f_{xx} = 2\sqrt{2} \quad f_{yy} = 2\sqrt{2} \quad f_{xy} = 0$$

$$D = (2\sqrt{2})(2\sqrt{2}) - 0^2 = 8 > 0$$

and $f_{xx} > 0 \Rightarrow$ local minimum.

- Critical point $\left(\frac{1}{e\sqrt{2}}, \frac{1}{e\sqrt{2}}\right)$

$$x=y=\frac{1}{e\sqrt{2}}$$

- Compute f_{xx}, f_{yy}, f_{xy} :-

$$f_{xx} = f_{yy} = -2e^2 \quad f_{xy} = -2e^2$$

$$D = (-2e^2)^2 - (-2e^2)^2 = 4e^4 - 4e^4 = 0$$

test fails.

22. $f(x, y) = (x - y)e^{x^2 - y^2}$

- finding the partial derivatives.

$$f(x, y) = (x - y)e^{x^2 - y^2}$$

$$\begin{aligned} f_x &= \frac{\partial}{\partial x} \left[(x - y)e^{x^2 - y^2} \right] = e^{x^2 - y^2} + (x - y) \cdot e^{x^2 - y^2} \cdot 2x \\ &= e^{x^2 - y^2} [1 + 2x(x - y)] \end{aligned}$$

$$f_y = \frac{\partial}{\partial y} \left[(x-y)e^{x^2-y^2} \right] = -e^{x^2-y^2} + (x-y) \cdot e^{x^2-y^2} \cdot (-2y)$$

$$f_y = e^{x^2-y^2} [-1 - 2y(x-y)]$$

- finding the critical points-
 $f_x = 0$

$$1 + 2x(x-y) = 0 \quad \text{--- 1}$$

set $f_y = 0$

$$-1 - 2y(x-y) = 0 \quad \text{--- 2}$$

solver- eq (i) & (ii)

$$1 + 2x^2 - 2xy = 0 \rightarrow 2x^2 - 2xy = -1 \quad \text{--- 1}$$

$$-1 - 2xy + 2y^2 = 0 \rightarrow -2xy + 2y^2 = 1 \quad \text{--- 2}$$

- add eq (i) & (ii)

$$2x^2 - 2xy - 2xy + 2y^2 = 0$$

$$2x^2 - 4xy + 2y^2 = 0$$

$$(x-y)^2 = 0$$

$$\boxed{y = x}$$

- substitute $x=y$ in eq (i):-

$$2x^2 - 2x \cdot x = -1$$

$\boxed{0 = -1}$ → No critical point when f_x and f_y are both zero.



In Exercises 29–32, determine the global extreme values of the function on the given set without using calculus.

30. $f(x, y) = 2x - y$, $0 \leq x \leq 1$, $0 \leq y \leq 3$

The function $f(x, y) = 2x - y$ is a linear function. This means that its extreme values will occur at the corners (vertices) of the rectangular region defined by the given inequalities.

- finding the corners:-

$$(0, 0), (1, 0), (0, 3), (1, 3)$$

- evaluating the function at corners:-

- $(0, 0) : f(0, 0) = 2(0) - 0 = 0$

$(1, 0) : = 2$

$(0, 3) = -3$

$(1, 3) = -1$

- Global maximum, the largest value is '2' which occurs at $(1, 0)$

- Global minimum:- smallest value is '-3' at (0,3)

33. Assumptions Matter Show that $f(x, y) = xy$ does not have a global minimum or a global maximum on the domain

$$D = \{(x, y) : 0 < x < 1, 0 < y < 1\}$$

Explain why this does not contradict Theorem 3.

THERE IS NO GLOBAL MAXIMUM:-

Let's assume there's a global maximum M . This means that $xy \leq M$ for all (x, y) in D .

However, since x and y can get arbitrarily close to 1, their product xy can get arbitrarily close to 1.

For any M less than 1, we can find x and y such that $xy > M$. For example, if $M = 0.99$, we can choose $x = 0.995$ and $y = 0.995$, then $xy = 0.990025 > 0.99$.

This means we can always find values of x and y in D such that xy is larger than our assumed maximum M . Therefore, there is no global maximum.

THERE IS NO GLOBAL MINIMUM:-

Similarly, let's assume there's a global minimum m . This means that $xy \geq m$ for all (x, y) in D .

However, since x and y can get arbitrarily close to 0, their product xy can get arbitrarily close to 0.

For any m greater than 0, we can find x and y such that $xy < m$. For example, if $m = 0.001$, we can choose $x = 0.01$ and $y = 0.01$, then $xy = 0.0001 < 0.001$.

This means we can always find values of x and y in D such that xy is smaller than our assumed minimum m . Therefore, there is no global minimum.

- Why it doesn't contradict with theorem 3:-

- If f is a continuous function on a closed domain then f attains both max and min values on D .
- In this case domain D is an open set, it is not closed because it doesn't include its boundaries. Therefore,

Theorem 3 doesn't apply for it.

35. Find the maximum of

$$f(x, y) = x + y - x^2 - y^2 - xy$$

on the square, $0 \leq x \leq 2$, $0 \leq y \leq 2$ (Figure 23).

(a) First, locate the critical point of f in the square, and evaluate f at this point.

(b) On the bottom edge of the square, $y = 0$ and $f(x, 0) = x - x^2$. Find the extreme values of f on the bottom edge.

(c) Find the extreme values of f on the remaining edges.

(d) Find the largest among the values computed in (a), (b), and (c).

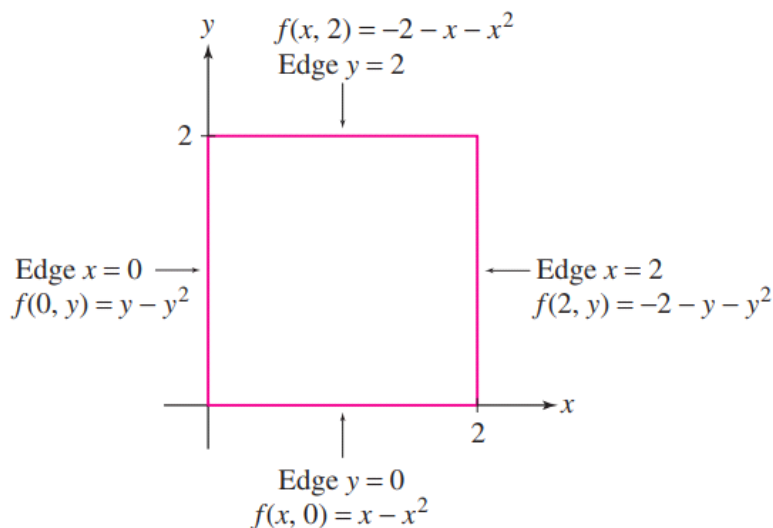


FIGURE 23 The function $f(x, y) = x + y - x^2 - y^2 - xy$ on the boundary segments of the square $0 \leq x \leq 2$, $0 \leq y \leq 2$.

- Compute the partial derivatives

$$f_x = \frac{\partial f}{\partial x} = 1 - 2x - y$$

$$f_x = \frac{\partial f}{\partial x} = 1 - 2y - x$$

- finding critical points

$$f_x = 0 \text{ and } f_y = 0$$

$$1 - 2x - y = 0$$

$$1 - 2y - x = 0$$

- Solve the system of equations:-

$$y = 1 - 2x$$

$$1 - 2(1 - 2x) - x = 0$$

$$1 - 2 + 4x - x = 0$$

$$\boxed{x = 1/3}$$

- Then $y = 1 - 2 \cdot 1/3 = 1/3$

- evaluate f at critical points:-

$$f\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{1}{3} + \frac{1}{3} - \left(\frac{1}{3}\right)^2 - \left(\frac{1}{3}\right)^2 - \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{3}$$

- b) finding extreme values:- ($y=0$)

- express f on the bottom edge:-

$$f(x, 0) = x - x^2$$

$$\frac{d}{dx} (x - x^2) = 1 - 2x = 0 \Rightarrow x = 1/2$$

- evaluate f at critical points :-

$$x = 1/2 :- f(1/2, 0) = \frac{1}{2} - \left(\frac{1}{4}\right) = \boxed{1/4}$$

at $x = 0 :-$

$$f(0, 0) = 0 - 0 = \boxed{0}$$

at $x = 2 :-$

$$f(2, 0) = 2 - 4 = \boxed{-2}$$

- extreme values on bottom edge :-

- Max : $1/4$ at $x = 1/2$

- Min : -2 at $x = 2$

c) Top edge ($y = 2$)

$$f(x, 2) = x + 2 - x^2 - 4 - 2x = -2 - x - x^2$$

- find the critical points:-

$$\frac{d}{dx} (-2 - x - x^2) = -1 - 2x = 0 \Rightarrow x = -1/2$$

- as $x = -1/2$ is outside the interval $[0, 2]$, so evaluate at boundaries:-

at $x=0$:-

$$f(0, 2) = -2 - 0 = -2 \quad \text{Max}$$

at $x=2$:-

$$f(2, 2) = -2 - 2 - 4 = -8 \quad \text{Min}$$

- left edge ($x=0$):-

$$f(0, y) = y - y^2$$

- critical points:-

$$\frac{d}{dy} (y - y^2) = 1 - 2y = 0 \quad y = 1/2$$

- evaluate :-

$$y = 1/2$$

$$f(0, 1/2) = 1/2 - (1/4) = 1/4 \quad \text{Max}$$

at $y=0$

$$f(0, 0) = 0$$

at $y=2$:-

$$f(0, 2) = 2 - 4 = -2 \quad \text{Min}$$

- right edge ($x=2$)

$$f(2, y) = 2 + y - 4 - y^2 - 2y = -2 - y - y^2$$

$$\frac{d}{dy} (-2 - y - y^2) = -1 - 2y = 0 \Rightarrow y = -1/2$$

as $y = -1/2$ is outside the interval, evaluate at boundary points: $[0, 2]$

at $y=0$:-

$$f(2, 0) = -2 - 0 = -2 \quad \text{Max}$$

at $y=2$

$$f(2, 2) = -2 - 2 - 4 = -8 \quad \text{Min}$$

d) find the largest amongst values:-

- $1/3$ critical point
- $1/4$ (bottom and left edges)
- -2 (various points)

max value of f on the square is " $1/3$ ".



In Exercises 37–45, determine the global extreme values of the function on the given domain.

43. $f(x, y) = x^2 + y^2 - 2x - 4y, \quad x \geq 0, \quad 0 \leq y \leq 3, \quad y \geq x$

- Compute the partial derivatives:-

$$f_x = \frac{\partial f}{\partial x} = 2x - 2$$

$$f_y = \frac{\partial f}{\partial y} = 2y - 4$$

- Critical points = $f_x = 0$ $f_y = 0$

$$2x - 2 = 0, x = 1$$

$$2y - 4 = 0, y = 2$$

- evaluate f at critical point.

$$f(1, 2) = (1)^2 + 2^2 - 2 - 4(2) = -5$$

- analyze the boundary of domain-

1- line $y = 0$ (where $0 \leq x \leq y = 0$ so $x = 0$)

$$f(0, 0) = 0 + 0 - 0 - 0 = 0$$

2- line $y = 3$ (where $0 \leq x \leq 3$)

$$f(x, 3) = x^2 + 9 - 2x - 12 = x^2 - 2x - 3$$

- critical points:-

$$\frac{d}{dx} (x^2 - 2x - 3) = 2x - 2 = 0$$

$$x = 1$$

- evaluate at $x=1$:-

$$f(1,3) = 1 - 2 - 3 = -4$$

- at $x=0$:-

$$f(0,3) = -3$$

- at $x=3$:-

$$f(3,3) = 0$$

3. at line $y=x$ (where $0 \leq x \leq 3$):-

$$f(x,x) = x^2 + x^2 - 2x - 4x = 2x^2 - 6x$$

$$\frac{d}{dx} (2x^2 - 6x) = 4x - 6 = 0$$

$$x = 3/2$$

- evaluate:

- at $x=3/2$:-

$$f(3/2, 3/2) = 2(9/4) - 9 = -9/2$$

at $x=0$:-

$$f(0,0) = 0$$

at $x=3$:-

$$f(3,3) = 0$$

4. line $x=0$ (where $0 \leq y \leq 3$):-

f on $x=0$ -

$$f(0,y) = y^2 - 4y$$

- critical points:-

$$\frac{d}{dy}(y^2 - 4y) \Rightarrow y = 2$$

• evaluate at:-

$$y = 2 \Rightarrow f(0,2) \rightarrow -4$$

$$y = 0 \Rightarrow f(0,0) \rightarrow 0$$

$$y = 3 \Rightarrow f(0,3) \rightarrow -3$$

• Compare all values:-

$$\text{critical point : } f(1,2) = -5$$

- Boundary:-

$$f(0,0) = 0$$

$$f(1,3) = -4$$

$$f(0,3) = -3$$

$$f(3,3) = 0$$

$$f(3/2, 3/2) = -9/2$$

$$f(0,2) = -4$$

• Global Max = -5 at $(1,2)$

Global Min = 0 at $(0,0), (3,3)$

50. Show that the rectangular box (including the top and bottom) with fixed volume $V = 27 \text{ m}^3$ and smallest possible surface area is a cube (Figure 25).

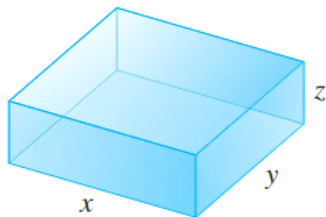


FIGURE 25 Rectangular box with sides x, y, z .

51.  Consider a rectangular box B that has a bottom and sides but no top and has minimal surface area among all boxes with fixed volume V .

(a) Do you think B is a cube as in the solution to Exercise 50? If not, how would its shape differ from a cube?

(b) Find the dimensions of B and compare with your response to (a).

a) No, B is not a cube, the height is smaller than the length and width.

b) we find that $x = y = (2V)^{1/3}$
 $z = (V/4)^{1/3}$

So, $V = 27$ we have.

$$x = y = (2 \times 27)^{1/3} = 3.78$$

$$z = (27/4)^{1/3} = 1.89$$

52. Find three positive numbers that sum to 150 with the largest possible product of the three.

$$x + y + z = 150$$

$$P = xyz \quad (P \text{ is maximum})$$

$$z = 150 - x - y$$

$$P = xy(150 - x - y)$$

$$P = 150xy - x^2y - xy^2$$

$$\bullet \quad \frac{\partial P}{\partial x} = 150y - 2xy - y^2 = 0$$

$$\bullet \quad \frac{\partial P}{\partial y} = 150x - x^2 - 2xy = 0$$

• Solve the system of equations:-

$$y(150 - 2x - y) = 0$$

$$x(150 - x - 2y) = 0$$

$$y = 150 - 2x$$

$$150 - x - 300 + 4x = 0$$

$$3x - 150 = 0$$

$$\boxed{x = 50}$$

$$\boxed{y = 50}$$

- check for max. -
second derivative test -

$$\bullet \frac{\partial^2 P}{\partial x^2} = -2y$$

$$\bullet \frac{\partial^2 P}{\partial y^2} = -2x$$

$$\bullet \frac{\partial^2 P}{\partial x \partial y} = 150 - 2x - 2y$$

At $(50, 50)$:-

$$\frac{\partial^2 P}{\partial x^2} = -100$$

$$\frac{\partial^2 P}{\partial x \partial y} = 150 - 100 - 100 = -50$$

$$\frac{\partial^2 P}{\partial y^2} = -100$$

$$\bullet D = (-100)(-100) - (-50)^2 = 7500$$

since $D > 0$ and $\frac{\partial^2 P}{\partial x^2} < 0$, we have max at $(50, 50)$

53. A 120-m long fence is to be cut into pieces to make three enclosures, each of which is square. How should the fence be cut up in order to minimize the total area enclosed by the fence?

let the side be x, y, z

- Perimeter = $4x, 4y, 4z$
- Total length of fence = $4x + 4y + 4z = 120$
Total Area = $x^2 + y^2 + z^2$
- Perimeter equation = $x + y + z = 30$
 $z = 30 - x - y$

$$A = x^2 + y^2 + (30 - x - y)^2$$

- Find partial derivatives

$$\frac{\partial A}{\partial x} = 2x + 2(30 - x - y)(-1) = 0$$

$$\frac{\partial A}{\partial y} = 2y + 2(30 - x - y)(-1) = 0$$

$$4x + 2y = 60$$

$$2x + 4y = 60$$

Solve:-

$$x = 10, y = 10$$

$$z = 10$$

- check for minimum:-

$$\frac{\partial^2 A}{\partial x^2} = 4 \quad \frac{\partial^2 A}{\partial y^2} = 4 \quad \frac{\partial^2 A}{\partial x \partial y} = 2$$

- Now $D = 4(4) - (2)^2 = 16 - 4 = 12$

since $D > 0$ and $\frac{\partial^2 A}{\partial x^2} > 0$ we have local

min at $(10, 10)$.

- calculate the fence lengths:-

$$u_x = 40 \text{ meters}$$

$$u_y = 40 \text{ meters}$$

$$u_z = 40 \text{ meters}$$

54. A box with a volume of 8 m^3 is to be constructed with a gold-plated top, silver-plated bottom, and copper-plated sides. If gold plate costs \$120 per square meter, silver plate costs \$40 per square meter, and copper plate costs \$10 per square meter, find the dimensions that will minimize the cost of the materials for the box.

$$V = 8 \text{ m}^3$$

let length = x , width = y , height = z

$$xyz = 8$$

• Now,

$$C = 120xy (\text{top}) + 40xy (\text{bottom}) + 10(2xz + 2yz) (\text{sides})$$

$$= 160xy + 20(x+y)z$$

$$\text{as } z = \frac{8}{xy}$$

$$C = 160xy + 20(x+y) \left(\frac{8}{xy} \right) = 160xy + \frac{160}{y} + \frac{160}{x}$$

$$C_x = 160y - \frac{160}{x^2} = 0 \quad y = 1/x^2$$

$$C_y = 160x - \frac{160}{y^2} = 0 \quad x = 1/y^2$$

$$x = \frac{1}{(1/x)^2} = x^4 \Rightarrow x = 1 \text{ m}, y = 1 \text{ m}, z = 8 \text{ m}$$

• final dimensions:- $1 \text{ m} \times 1 \text{ m} \times 8 \text{ m}$

55. Find the maximum volume of a cylindrical soda can such that the sum of its height and its circumference is 120 cm.

• let radius = r , height = h .

$$V = \pi r^2 h$$

$$S = 2\pi r^2 + 2\pi r h$$

$$h = \frac{S - 2\pi r^2}{2\pi r}$$

- substitute into V:-

$$V = \pi r^2 \left(\frac{S - 2\pi r^2}{2\pi r} \right) = \frac{S r}{2} - \pi r^3$$

- Critical points:-

$$\frac{dV}{dr} = \frac{S}{2} - 3\pi r^2 = 0 \quad r = \sqrt{\frac{S}{6\pi}}$$

$$h = \frac{S - 2\pi \left(\frac{S}{6\pi} \right)}{2\pi \sqrt{\left(\frac{S}{6\pi} \right)}} = 2 \sqrt{\frac{S}{6\pi}}$$

ratio:-

$$\boxed{\frac{h}{r} = 2}$$

- final dimensions:-

$$r = \sqrt{\frac{S}{6\pi}}, \quad h = 2\sqrt{\frac{S}{6\pi}}$$